

# WASP-12b

R-0458

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-12\_210412 · created 2026-07-19T19:48:16+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                  |
|---------------------------------------------------|----------------------------------|
| Mid-Transit Time (Tmid)                           | 2459316.6787 +/- 0.00083 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.155 +/- 0.045                  |
| Transit depth (Rp/Rs)^2                           | 2.4 +/- 1.38 [%]                 |
| Orbital Inclination (inc)                         | 76.94 +/- 2.87                   |
| Airmass coefficient 1 (a1)                        | 1.31 +/- 0.14                    |
| Airmass coefficient 2 (a2)                        | -0.056 +/- 0.03                  |
| Scatter in the residuals of the lightcurve fit is | 10.53 %                          |
| Best Comparison Star                              | #2 - [210, 354]                  |
| Optimal Aperture                                  | 1.54                             |
| Optimal Annulus                                   | 10.07                            |
| Transit Duration (day)                            | 0.095 +/- 0.01                   |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.155 ± 0.045 vs 0.1178 (deviation 0.61σ) |
| Duration fit vs published | 0.095 d vs 0.125 d (deviation 2.21σ)      |
| Mid-time O–C              | -1.8 min                                  |
| Depth SNR                 | 2.5                                       |
| Residual scatter          | 10.53 %                                   |

### Light curve

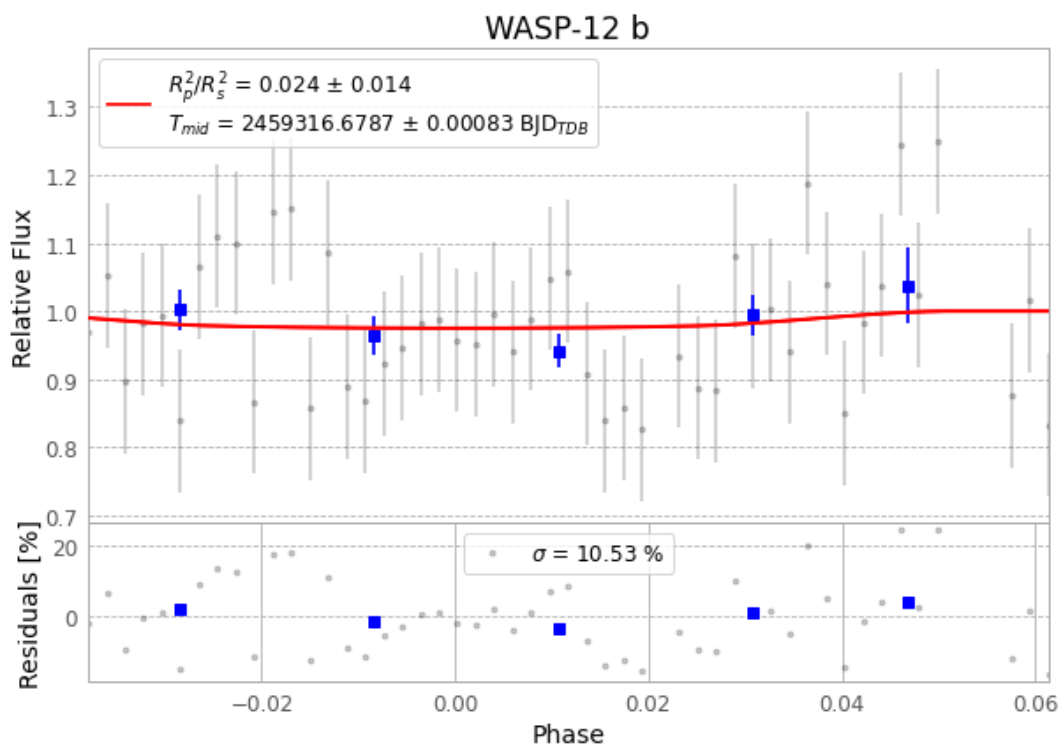


Figure 1 — FinalLightCurve\_WASP-12 b\_2021-04-11.png (SHA-256 4fb6373b797abc58...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [360, 223], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 b426c00d47076613...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 39d9b80

## Data provenance

74 FITS frames (49 MB), WASP-12210412031815.FITS → WASP-12210412065712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-210412.log (bccda2036f5c...), FinalLightCurve\_WASP-12 b\_2021-04-11.png (4fb6373b797a...), FinalLightCurve\_WASP-12 b\_2021-04-11.csv (19b758fda314...)

## Compute resources

Wall time 205.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 19:48Z.